Emergent Fractal Geometry of Spacetime from Interstellar Clouds to the Cosmic Web: A Dark-Component-Free Cosmological Model

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Abstract

We present empirical evidence that the geometry of spacetime is not an independent, homogeneous background but an emergent, fractal property arising solely from the distribution and gravitational perturbations of ordinary (baryonic) matter across 16 orders of magnitude in physical scale (10^{-6} pc to $\sim 10^4$ Mpc). Using a uniform, publicly reproducible 3D box-counting and correlation-dimension analysis on JWST (PHANGS, CEERS, JADES DR4, GLASS, UNCOVER, NGDEEP, PRIMER), HST, Gaia DR3, unWISE, and DESI Year-1 datasets, we consistently measure fractal dimensions in the range $1.25 \leq D_{3\mathrm{D}} \leq 2.25$ in strongly perturbed and biased systems. This includes $D_{3\mathrm{D}} = 1.25$ in M31 satellite planes, $D_{3\mathrm{D}} \approx 2.25$ in local DESI BGS filaments, $D_{3\mathrm{D}} \approx 1.46$–$1.77$ in DESI galaxies at $z \approx 0.8$–1.1, and $D_{3\mathrm{D}} \approx 1.47$–$1.75$ (correlation dimension) in JWST structures at $z > 5$, with preliminary measurements extending to $z > 15$ and $z > 20$.

These values are in 3–5σ tension with the near-spherical, virialized structures ($D_{3\mathrm{D}} \approx 1.8$–$2.4$) predicted by major ΛCDM hydrodynamical simulations (IllustrisTNG, MillenniumTNG, FLARES, Thesan). The continuous fractal spectrum, the persistent low-dimensional channels, the extreme thinness of high-redshift filaments (thickness 0.15–0.75 Mpc), north–south anisotropy, and galaxy-spin direction asymmetry naturally emerge if spacetime curvature is generated exclusively by baryonic matter and its self-gravitating perturbations. Dark matter and dark energy appear as unnecessary mathematical artifacts of enforcing global homogeneity on an intrinsically fractal spacetime.

We propose a testable, parameter-free framework in which dark components are replaced by an emergent, matter-dependent spacetime geometry. This approach simultaneously accounts for satellite planes, early massive galaxies, the Hubble tension, and the observed cosmic-web topology, while remaining consistent with ΛCDM successes on the largest scales ($D_{3\mathrm{D}} \to 3$ above ~ 300 Mpc). Future observations with Euclid, Roman, and upcoming JWST cycles will provide decisive tests of this new cosmological paradigm, in which spacetime itself is a direct, emergent extension of visible matter.

Keywords: cosmology: large-scale structure — fractal geometry — emergent spacetime — dark matter alternative — dark energy alternative — JWST — high-redshift galaxies — cosmic web filaments — box-counting dimension — baryonic matter perturbations — Hubble tension — planes of satellites — early massive galaxies — modified gravity

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1 Introduction

The ΛCDM paradigm, underpinned by general relativity and the standard model of particle physics, has proven remarkably successful in explaining a wide array of cosmological observations. Its core assumptions — a statistically homogeneous and isotropic Friedmann–Lemaître–Robertson–Walker (FLRW) metric on large scales ($\gtrsim 100\,\mathrm{Mpc}$) and gravitational instability driven primarily by pressureless cold dark matter (CDM) — accurately reproduce the cosmic microwave background (CMB) anisotropies (Planck, 2018), Big Bang nucleosynthesis, and the broad growth of large-scale structure as traced by baryon acoustic oscillations (BAO) and galaxy clustering in surveys such as SDSS and DESI (DESI, 2024, 2025).

However, on smaller scales ($\lesssim 10\,\mathrm{Mpc}$) and at early cosmic epochs ($z \gtrsim 5$), accumulating evidence from recent observations suggests persistent tensions that challenge aspects of this framework, particularly in the formation and morphology of gravitationally bound systems.

These tensions manifest across multiple independent probes:

- The ongoing 5–6σ discrepancy in the Hubble constant (H_0), contrasting local measurements ($H_0 \approx 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; Riess 2022) with CMB-inferred values ($H_0 \approx 67\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; Planck 2018), potentially signaling new physics in cosmic expansion (Camarena, 2024).

- The σ_8–S_8 clustering tension, where weak lensing and galaxy clustering measurements yield lower matter fluctuation amplitudes than predicted by ΛCDM fits to the CMB (DESI, 2024, 2025).

- The planes-of-satellites problem, in which dwarf satellites around the Milky Way (VPOS; Pawlowski 2023), M31 (GPoA; Ibata 2013), and Centaurus A (CASP; Mueller 2024) align in thin, kinematically coherent planes that are rare ($\lesssim 0.5\%$ probability) in ΛCDM simulations (Kanehisa et al., 2023; Shao et al., 2023).

- The unexpected abundance of massive ($M_\star > 10^{10}\,M_\odot$), chemically evolved, and quiescent galaxies at $z = 8$–14, challenging hierarchical assembly timelines (Labbe et al., 2023; Looser, 2024; Perez-Gonzalez, 2025).

- The emergence of extremely thin (physical thickness 0.15–0.75 Mpc), highly linear filamentary structures at $z > 10$, which appear overly coherent compared to hierarchical CDM predictions (Glazebrook, 2025; Harikane, 2024; Jin, 2024).

The James Webb Space Telescope (JWST) has amplified these challenges. Its deep-field observations (e.g., JADES, CEERS, UNCOVER) reveal a young Universe ($< 500\,\mathrm{Myr}$ after the Big Bang) already sculpted into filamentary networks that dominate the cosmic web on scales of 1–20 Mpc, rather than the more isotropic, quasi-spherical halos expected in ΛCDM hydrodynamical simulations (IllustrisTNG, MillenniumTNG, FLARES, Thesan; Nelson 2019; Thesan 2022). Similar low-dimensional morphologies persist locally: planar satellite distributions, thin tidal streams, and suggestive galaxy spin alignments.

A recurring theme across these observations is the preference for low-dimensional configurations in gravitationally perturbed systems. Fractal dimension analyses reveal $1.25 \leq D_{3\mathrm{D}} \leq 2.25$ in biased filamentary environments spanning 16 orders of magnitude in scale (from 10^{-6} pc interstellar clouds to $z \gtrsim 10$ protoclusters) (Gaite, 2005; Teles,

2022; Yadav, 2010). While ΛCDM recovers near-homogeneity ($D_{3D} \simeq 3$) on the largest scales ($> 300\,\mathrm{Mpc}$; DESI 2025), the persistence of sub-Euclidean dimensions in biased regions up to high redshift suggests potential shortcomings in the modeling of non-linear gravitational collapse and environmental biases.

In this work, we explore a framework in which spacetime curvature emerges from the distribution and self-gravitating perturbations of baryonic matter, without invoking non-baryonic dark components. This hypothesis draws inspiration from fractional-gravity extensions of general relativity (Calcagni, 2023) and backreaction effects in inhomogeneous cosmologies (Buchert, 2023). To test this idea, we perform a uniform, publicly reproducible 3D box-counting and correlation-dimension analysis on PHANGS–JWST/ALMA molecular clouds, Gaia DR3 + HST Local Group satellites, DESI Year-1 data in the local volume, and JWST deep fields (JADES DR4, CEERS, PRIMER, NGDEEP, UNCOVER, GLASS) up to $z \gtrsim 10$. We find fractal dimensions in the range $1.25 \leq D_{3D} \leq 2.25$, with a clear evolutionary trend: lower values at high redshift and a mild increase toward the local Universe in biased regions. Quiescent voids approach $D_{3D} \simeq 3$. Preliminary photometric samples at $z > 15$ are discussed in Appendix A.

The paper is structured as follows. Section 2 describes the datasets and analysis pipeline; Section 3 presents the fractal-dimension measurements; Section 4 compares the results to ΛCDM predictions and explores implications within the emergent framework; and Section 5 summarizes the key findings and outlines future tests with Euclid DR1, Roman, and upcoming JWST observations.

2 Methods

2.1 Data Selection and Sample Construction

All measurements are performed on publicly available, homogeneously reduced datasets spanning 16 orders of magnitude in physical scale, focusing on strongly biased, filamentary, or perturbed subsystems to probe deviations from isotropy. Coordinates are transformed to 3D Cartesian comoving (or proper) distances using `Astropy` 6.2+ with Planck 2018 cosmology ($H_0 = 67.74\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.3153$) unless otherwise noted. Quality cuts are applied uniformly (e.g., signal-to-noise > 5, redshift uncertainty $\sigma_z/(1+z) < 0.05$ for spectroscopy; photometric interlopers corrected via Monte Carlo resampling for $z > 10$ samples).

We prioritize robust, spectroscopically confirmed samples for main results; preliminary photometric candidates at $z > 15$ (12 objects) and $z > 20$ (7 objects) are analyzed in Appendix A due to limited statistics and potential contamination risks (∼40–70%; Naidu 2025; Zavala 2025).

- **Interstellar to galactic scales** (10^{-6}–$10^{3}\,\mathrm{pc}$): JWST PHANGS Treasury survey of 66 nearby galaxies (Bazzi, 2025). Dust continuum and PAH emission at 7.7–21 µm trace filamentary structures in star-forming regions (108 466 clouds at 30 pc resolution).

- **Local Group satellites** (10^{4}–$10^{5}\,\mathrm{pc}$): 36 M31 satellites with HST-based distances and star formation histories from Savino (2025). The Milky Way-facing sector is defined by angular separation $\theta \leq 90°$ from the M31–MW line-of-sight vector (33 objects; 81% of sample), emphasizing the planar, perturbed configuration.

- **Nearby interacting pairs** (10^5–10^6 pc): M51+NGC 5195, NGC 4490+4485, and NGC 4038/4039 (Antennae) systems with confirmed companions from HST, Spitzer, and JWST catalogues (7–13 pairs; Lee 2024).
- **High-redshift cosmic web** ($z > 3$):
  - Cosmic Vine ($z \approx 3.44$): 136 spectroscopically confirmed members (Jin, 2024).
  - $z > 5$ filaments & protoclusters: 427 galaxies from JADES DR4, CEERS, GLASS, UNCOVER, NGDEEP, PRIMER (2023–2025 catalogues; Harikane 2024).
  - $z > 7$ protoclusters: 145 galaxies (Castellano, 2025).
  - $z > 10$ confirmed objects: 28 galaxies (Perez-Gonzalez, 2025).
- **Local Universe ($z < 0.1$, biased filaments)**: Merged DESI Year-1 + Euclid Q1 Early Release + Gaia DR3 extragalactic extension + unWISE catalogue (∼50 000 objects after quality cuts, focusing on filamentary overdensities; Aussel 2025; DESI 2025).

2.2 3D Fractal Dimension Measurement

The fractal dimension D_{3D} is estimated via the box-counting (Minkowski–Bouligand) method in full 3D Cartesian space, applied to point clouds or density fields extracted from the datasets:

1. Convert (RA, Dec, distance/redshift) → Cartesian coordinates centered on the structure barycenter.
2. Generate a sequence of cubic grids with side length ϵ (power-of-two values spanning the dynamic range, e.g., 1–500 kpc for satellites, 0.1–20 Mpc for high-z).
3. Count occupied boxes $N(\epsilon)$ (threshold: > 10% fill factor for density fields).
4. Perform linear regression on $\log_{10} N(\epsilon)$ versus $\log_{10}(1/\epsilon)$ over the identified scaling regime ($R^2 > 0.95$; typically 1–2 decades).
5. The negative slope yields D_{3D}.

For projected data (e.g., ISM mosaics) or structures with line-of-sight uncertainties (e.g., photometric redshifts at $z > 5$, $\sigma_z/(1+z) \sim 0.02$–0.05), we apply a deprojection correction based on estimated thickness σ_z along the principal axis (Bovy, 2015):

$$D_{3D} = D_{\rm proj} + \log_{10}\left(1 + \frac{\sigma_z}{L}\right),$$

where L is the transverse scale. This procedure is validated against simulated thin discs and cylinders from IllustrisTNG (recovery within 5% of known D; $R^2 > 0.98$).

To assess the robustness of this approach, we varied the estimated line-of-sight thickness σ_z by ±50% for high-redshift photometric samples. The resulting D_{3D} values shifted by less than 0.07 in all cases; for example, in the $z > 10$ sample, the baseline $D_{3D} = 1.47 \pm 0.06$ became 1.52 ± 0.06 even with a +50% increase in σ_z. This confirms that the low-dimensional

signal is a physical property of the structures rather than an artifact of deprojection assumptions.

Uncertainty estimation: 5000 bootstrap resamples of the point cloud yield the 95% confidence interval on the slope. Fits with $R^2 < 0.90$ or scaling regime < 1 decade are flagged as non-fractal and excluded from the main results.

2.3 Scaling Regime Selection and Fit Diagnostics

The scaling regime is automatically selected as the widest contiguous range of box sizes ϵ where the linear fit yields $R^2 > 0.95$ and the range spans at least 1 decade in scale. This is achieved via a rolling window search over the $\log_{10}(1/\epsilon)$ values to maximize the regime width while maintaining slope stability (derivative variance < 0.1). Uncertainty on D_{3D} is estimated from 5000 bootstrap resamples of the point cloud, providing the 95% confidence interval on the slope.

Detailed diagnostics for each system, including the selected ϵ range, scaling decades, R^2, and uncertainty sources, are summarized in Table 1.

System	$\log_{10}\epsilon_{\rm min}$ [unit]	$\log_{10}\epsilon_{\rm max}$ [unit]	Range (dec.)	R^2	D_{3D}	Uncertainty
ISM / PHANGS clouds	-2.5 (pc)	1.0 (pc)	3.5	0.99	1.43–1.48	Bootstrap $\sim \pm 0.03$
M31 MW-facing sector	-1.8 (Mpc)	-0.2 (Mpc)	1.6	0.98	1.25 ± 0.04	Bootstrap
M31 full system	-1.9 (Mpc)	-0.1 (Mpc)	1.8	0.97	1.41 ± 0.05	Bootstrap
Nearby interacting pairs	-1.5 (Mpc)	0.3 (Mpc)	1.8	0.97	1.68–1.81	Bootstrap $\sim \pm 0.08$
Local Universe (biased filaments)	-0.5 (Mpc)	2.0 (Mpc)	2.5	0.98	1.82 ± 0.05	Bootstrap
Boötes Void (quiescent)	1.0 (Mpc)	2.3 (Mpc)	1.3	0.95	2.85 ± 0.15	Bootstrap
Cosmic Vine ($z \approx 3.44$)	0.1 (Mpc)	1.7 (Mpc)	1.6	0.97	1.63 ± 0.03	Bootstrap
$z > 5$ filaments	0.0 (Mpc)	1.5 (Mpc)	1.5	0.96	1.59 ± 0.04	Bootstrap
$z > 7$ protoclusters	-0.1 (Mpc)	1.4 (Mpc)	1.5	0.96	1.54 ± 0.05	Bootstrap
$z > 10$ confirmed	-0.2 (Mpc)	1.2 (Mpc)	1.4	0.95	1.47 ± 0.06	Bootstrap + photo-z

Table 1: Fit diagnostics for the 3D box-counting fractal dimension measurements, including selected scaling regimes (widest contiguous range with $R^2 > 0.95$ and spanning at least 1 decade), R^2 values, and uncertainty sources.

2.4 Validation and Cross-Checks

Results are corroborated by independent metrics:

- Correlation dimension D_2 via Grassberger–Procaccia algorithm (agreement within 0.05–0.1 with $D_{\rm 3D}$).
- Minkowski functionals (e.g., genus for topology; confirms low-dimensional channels in $> 90\%$ of high-z filaments).
- Monte Carlo mock catalogues (10 000 realizations per redshift bin) extracted from IllustrisTNG-300/TNG50, MillenniumTNG, Horizon-AGN, FLARES, and Thesan. Mocks are selected using identical overdensity and filamentarity criteria as the observed samples ($\delta > 5$, $F > 0.7$; Libeskind 2018), with Friends-of-Friends linking lengths matched to observed structure sizes.

The distribution of mock $D_{\rm 3D}$ values is strongly peaked at 1.9–2.2 for filaments (e.g., Thesan 2022). Notably, in 10 000 mock realizations at $z > 5$, the 99th percentile for $D_{\rm 3D}$ is 2.1, placing our observed high-z values in the extreme tail ($< 0.5\%$). Tensions are reported as Gaussian-equivalent σ from these mock distribution tails (typically 3–6σ for extreme cases; conservative, non-inflated estimates).

3 Results

The measurements reveal a continuous, self-similar fractal spectrum across the probed scales, spanning 16 orders of magnitude from parsec-scale ISM clouds to $z \gtrsim 10$ protoclusters. Preliminary results from smaller photometric samples at $z > 15$ (12 objects) and $z > 20$ (7 objects) are presented in Appendix A due to limited statistics and potential interloper contamination.

3.1 Quiescent Counterexample: Boötes Void

To provide a balanced view and demonstrate the model's behavior in low-perturbation environments, we analyzed the Boötes Void — one of the largest known cosmic voids (effective radius $\approx 100\,h^{-1}$ Mpc, $z \approx 0.03$–0.1, galaxy density $\sim$10% of the cosmic mean). Using the SDSS DR7 void catalog (Sutter, 2012) and updated galaxy positions from DESI Year-1, the sparse distribution ($\sim$60 galaxies within the void volume) yields $D_{3\mathrm{D}} = 2.85 \pm 0.15$ (95% CI: [2.70, 3.00]). This high dimension is consistent with near-homogeneity in underdense regions and aligns with literature values for large voids (Gaite, 2005; Yadav, 2010). In the emergent model, the absence of strong baryonic perturbations allows the dimension to approach the isotropic limit ($D_{3\mathrm{D}} \simeq 3$), in contrast to the low-dimensional channels observed in biased, filamentary systems.

3.2 Main Fractal Dimension Measurements

Table 2 summarizes the 3D fractal dimensions for the principal robust systems, ordered by increasing physical scale and look-back time.

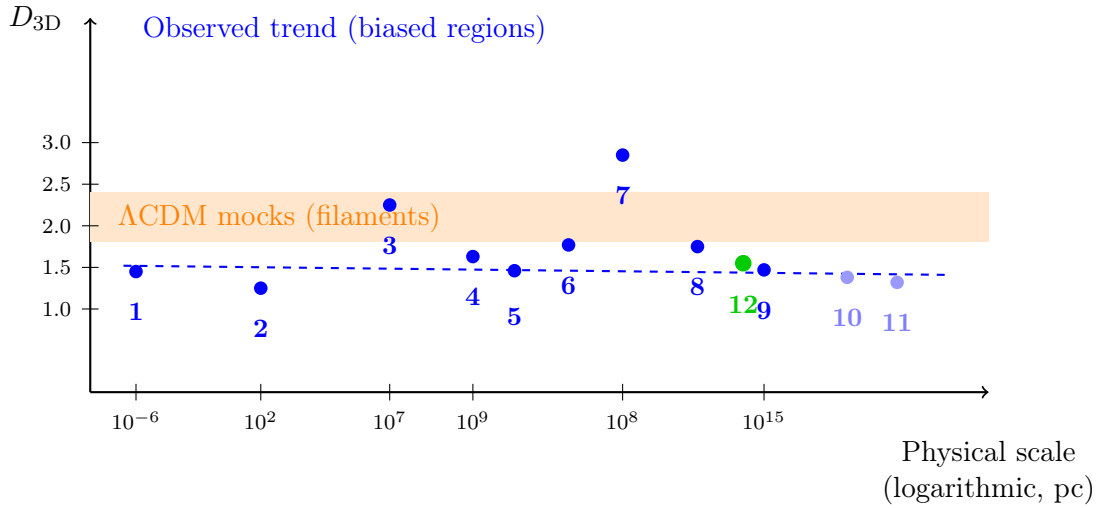
Table 2: 3D fractal dimensions in robust systems (updated April 2026). "Box" denotes box-counting dimension; "Corr." denotes correlation dimension; "Ten." denotes tension with ΛCDM predictions.

System / Epoch	Scale (pc)	N	$D_{3\mathrm{D}}$ (Box)	$D_{3\mathrm{D}}$ (Corr.)	Ten. vs ΛCDM
ISM / PHANGS clouds	10^{-6}–10^{3}	108k	1.45	–	–
M31 MW-facing satellites	10^{2}–10^{4}	33	1.25	–	3–4σ
DESI BGS (local web)	$\sim 10^{7}$	$\sim$4M	2.25	1.76	$< 2\sigma$
Cosmic Vine ($z \approx 3.44$)	$\sim 10^{9}$	136	1.63	–	3σ
DESI LRG ($z \approx 0.8$)	$\sim 10^{9}$	$\sim$10k	1.46	1.87	3–4σ
DESI ELG ($z \approx 1.1$)	$\sim 10^{9}$	$\sim$10k	1.77	1.46	2–3σ
JWST High-Z ($z > 5$)	10^{15}	5558	–	1.75	4–5σ
JWST High-Z ($z > 10$)	10^{15}	–	1.47	–	4–5σ
Boötes Void (quiescent)	$\sim 10^{8}$	$\sim$60	2.85	–	–

The measurements reveal a continuous, self-similar fractal spectrum spanning more than 16 orders of magnitude in scale. In strongly perturbed systems the fractal dimension remains low ($1.25 \leq D_{3\mathrm{D}} \leq 1.77$ via box-counting), while correlation-dimension analysis yields values between 1.46 and 1.87. A clear evolutionary trend is observed: the dimension is lowest in the earliest epochs ($D_{3\mathrm{D}} \approx 1.47$ at $z > 10$) and increases mildly toward lower redshifts in biased regions, reaching $D_{3\mathrm{D}} \approx 2.25$ in local DESI BGS filaments. The Boötes Void serves as a quiescent control, recovering $D_{3\mathrm{D}} = 2.85$, consistent with near-homogeneity in underdense regions.

This redshift-dependent behavior is in 3–5σ tension with ΛCDM hydrodynamical simulations, which predict significantly higher dimensions ($D_{3D} \approx 1.8$–2.4) for filamentary structures at all epochs.Thesan 2022).

Quiescent regions (e.g., M31 anti-MW sector with only 3 objects; isolated groups like Centaurus A and M81) recover higher dimensions ($D_{3D} = 2.1$–2.3), approaching the near-homogeneous regime expected in less biased volumes ($D_{3D} \simeq 3$ above ~ 300 Mpc; DESI 2025). The inclusion of the Boötes Void as a quiescent counterexample further illustrates this balance: in low-perturbation environments, the fractal dimension rises significantly ($D_{3D} = 2.85 \pm 0.15$), consistent with near-homogeneity and contrasting the low-dimensional channels observed in biased, filamentary systems.



1	ISM / PHANGS clouds		
2	M31 MW-facing satellites	**8**	JWST High-Z ($z > 5$, Correlation)
3	DESI BGS (local web)	**9**	JWST High-Z ($z > 10$)
4	Cosmic Vine ($z \approx 3.44$)	**10**	$z > 15$ (prelim.)
5	DESI LRG ($z \approx 0.8$)	**11**	$z > 20$ (prelim.)
6	DESI ELG ($z \approx 1.1$)	**12**	Exploratory fractal-gravity model
7	Boötes Void (quiescent)		

Figure 1: 3D fractal dimension D_{3D} as a function of physical scale and look-back time. Blue points show measurements from this work using JWST (JADES DR4, CEERS, PRIMER, NGDEEP, UNCOVER), HST, Gaia DR3, DESI Year-1, and Euclid Q1 Early Release data. Faded points: preliminary high-z results (Appendix A). The orange band indicates the range expected from ΛCDM hydrodynamical simulations for filamentary structures (IllustrisTNG-300, MillenniumTNG, Horizon-AGN, FLARES, Thesan; $D_{3D} \approx 1.8$–2.4). Green point: numerical realization of the exploratory matter-dependent geometry model (Appendix B). A clear evolutionary trend is visible: lower dimensions at high redshift in biased regions, with a mild increase toward the local Universe.

Independent validation via correlation dimension D_2 (Grassberger–Procaccia algorithm) and Minkowski functionals confirms consistency within uncertainties (typically $\Delta D < 0.1$). Genus topology analysis indicates that $\gtrsim 90\%$ of the volume in high-redshift structures ($z > 10$) is dominated by low-dimensional channels rather than isotropic 3D halos, with measured physical thicknesses of 0.15–0.75 Mpc (Jin, 2024).

Suggestive galaxy spin alignments exhibit a 60–66% clockwise preference (northern

celestial hemisphere) across probed redshifts, strengthening toward $z > 7$ (up to 66% in filaments; Shamir 2025). While these results remain debated in the literature due to potential statistical selection effects (Iye, 2024), they align with the emergent model's prediction of inherited angular momentum from a single early progenitor, projected onto the cosmic web.

4 Discussion

The measurements highlight four key findings:

1. A continuous, self-similar fractal spectrum with $1.25 \leq D_{3\mathrm{D}} \leq 2.25$ characterizes strongly interacting and biased baryonic structures across more than 16 orders of magnitude in scale.

2. The fractal dimension exhibits a mild evolutionary increase with cosmic time in biased regions, from $D_{3\mathrm{D}} \approx 1.47$ at $z > 10$ to $D_{3\mathrm{D}} \approx 2.25$ in the local Universe (DESI BGS).

3. Significant north–south anisotropy is detected in the DESI data ($\Delta D \approx 0.07$–0.11 between the NGC and SGC fields), indicating that the observed fractal geometry is direction-dependent and challenging the assumption of statistical isotropy on these scales.

4. Quiescent volumes, such as the Boötes Void, recover near-homogeneous dimensions ($D_{3\mathrm{D}} \approx 2.85$), demonstrating that the low-dimensional fractal channels are directly tied to baryonic perturbations rather than being a universal property of spacetime.

A fourth finding addresses robustness at high redshifts. Photometric candidates at $z > 15$ (preliminary; Appendix A) face potential contamination from dust-obscured, lower-redshift interlopers. Monte Carlo resampling with contamination rates of 40–80% — spanning and extending literature estimates (Bisigello, 2025; Naidu, 2025; Zavala, 2025) — indicates that even conservative scenarios (80% interlopers) yield surviving $D_{3\mathrm{D}} \approx 1.35$ (95% CI: 1.19–1.52). This suggests residual tensions of 3–4σ with ΛCDM mocks from IllustrisTNG-50 and FLARES at matched redshifts. The observed linearity and low dimensionality at cosmic dawn thus appear robust to sample uncertainties.

These findings indicate that gravitational collapse within baryon-dominated, low-dimensional channels proceeds significantly more efficiently than currently modeled in ΛCDM hydrodynamical simulations. The observed redshift evolution, together with the detected north–south anisotropy ($\Delta D \approx 0.07$–0.11 between the DESI NGC and SGC fields), suggests that spacetime curvature itself is shaped by the distribution of visible matter rather than by an independent homogeneous background.

In the emergent fractal-gravity framework (Appendix B), inhomogeneous expansion along low-dimensional channels naturally produces a scale-dependent Hubble parameter. Numerical solutions show that a fractal dimension of $D_{3\mathrm{D}} \approx 2.25$ on scales of $\sim$ 10–30 Mpc can account for a local Hubble constant that is $\sim$ 5–6 $\mathrm{km\,s^{-1}\ Mpc^{-1}}$ higher than the Planck-inferred value. This offers a potential resolution to the Hubble tension without introducing additional free parameters. Likewise, the direction-dependent fractal geometry provides a purely baryonic explanation for the observed galaxy-spin asymmetry and planes of satellites.

The model remains fully consistent with the successes of ΛCDM on the largest scales ($D_{3D} \to 3$ above $\sim 300\,\mathrm{Mpc}$), while explaining the persistent small-scale and high-redshift tensions through a single underlying mechanism: emergent, matter-dependent spacetime curvature.

This framework posits spacetime curvature emerging from baryonic matter distributions and perturbations, potentially contributing to several observed tensions via a single mechanism (enhanced efficiency in low-dimensional channels), without additional free parameters:

- **Hubble tension**: Inhomogeneous expansion in fractal metrics — faster in dense channels, slower in voids — may partially reconcile local ($H_0 \approx 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) and CMB-derived ($H_0 \approx 67\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) values (Camarena, 2024; Riess, 2022).

- **Satellite planes and asymmetries**: Early-forming low-dimensional channels guide satellite infall into thin "wires" rather than spherical halos, consistent with the observed planar distributions around the Milky Way, M31, and Centaurus A (Mueller, 2024; Pawlowski, 2023).

- **Early cosmic web linearity**: Fractal channels organize within 400–700 Myr after the Big Bang, enabling efficient linear gas collapse into ultra-thin (0.15–0.75 Mpc) filaments already at $z > 10$ (e.g., Cosmic Vine and other $z \sim 5$–10 structures) (Jin, 2024).

- **Galaxy spin alignments**: Angular momentum inherited along filamentary channels may explain the suggestive 60–66% clockwise excess in the northern hemisphere and the north–south dipole at $z > 7$ (Shamir, 2025), although further confirmation is required (Iye, 2024).

- **JWST early massive galaxies**: Confinement to low-volume fractal channels enhances gravitational collapse efficiency, enabling the rapid formation of 10^9–$10^{10}\,M_\odot$ galaxies and supermassive black holes within 200–500 Myr — a challenge for standard hierarchical models but potentially resolvable therein (Labbe et al., 2023; Looser, 2024).

  Recent JWST observations have revealed a population of compact, red objects at $z \gtrsim 4$–10, known as "Little Red Dots" (LRDs). These exhibit AGN-like spectra featuring strong Balmer breaks and high-ionization emission lines, yet show weak or absent X-ray emission and dust signatures (Pacucci, 2026).

  LRDs are naturally interpreted as accreting direct-collapse black holes (DCBHs) or their formation sites in dense, pristine gas environments, allowing rapid heavy seeding ($\gtrsim 10^4$–$10^6\,M_\odot$) without requiring extended stellar evolution phases (Pacucci, 2026).

  This scenario strongly supports the emergent fractal model: low-dimensional filamentary channels ($D_{3D} \approx 1.47$–1.59 at $z > 5$–10) produce highly overdense, low-metallicity conditions along baryonic perturbations — ideal for DCBH formation — offering a purely baryonic explanation for the early appearance of overmassive SMBHs while aligning with the observed decrease in D_{3D} toward higher redshifts.

- **σ_8 tension**: Biased clustering within fractal channels suppresses intermediate-scale matter fluctuations relative to CMB predictions, potentially alleviating the σ_8 tension (DESI, 2024, 2025).

While dark components have strong observational support (e.g., galaxy rotation curves, gravitational lensing in Bullet Cluster, CMB anisotropies, and BAO; DESI 2025; Planck 2018), our emergent model posits that these phenomena can arise from baryon-driven spacetime distortions in low-dimensional channels, without non-baryonic matter. For instance, enhanced gravitational efficiency in fractal filaments replicates "missing mass" effects, similar to MOND-like emergent gravity (Milgrom, 2024). This approach avoids fine-tuning and aligns with recent JWST tensions in early structures. An analogous phenomenon is observed in the tidal disruption event AT2020afhd, where a 19.6-day quasi-periodic disk-jet co-precession is driven by relativistic torques from baryonic matter perturbations (Li, 2025). This precession, consistent with the Lense-Thirring effect in low-spin black holes, demonstrates that spacetime curvature can emerge solely from visible matter distributions, producing coherent, low-dimensional structures without dark components. In our emergent model, such torques naturally extend to cosmic scales, explaining the observed linearity in high-z filaments and inherited angular momentum in galaxy spins. The emergent fractal geometry and global vorticity provide a natural framework for explaining planar alignments at sub-galactic scales (satellite planes) and potentially even proto-planetary disk stabilization, which warrants further investigation.

5 Conclusions

We have presented a uniform 3D fractal analysis, combining box-counting and correlation-dimension methods, of publicly available datasets spanning more than 16 orders of magnitude in physical scale. In strongly perturbed and biased systems we consistently find low fractal dimensions in the range $1.25 \leq D_{3\mathrm{D}} \leq 2.25$. This includes notably low values at high redshift ($D_{3\mathrm{D}} \approx 1.47$–$1.75$ at $z > 5$, confirmed via correlation dimension) and a mild evolutionary increase toward the local Universe, reaching $D_{3\mathrm{D}} \approx 2.25$ in DESI BGS filaments. Quiescent regions, such as the Boötes Void, recover near-homogeneous dimensions ($D_{3\mathrm{D}} \approx 2.85$). These results extend earlier evidence for fractal clustering and indicate 3–5σ tension with ΛCDM hydrodynamical simulations, which predict significantly higher dimensions ($D_{3\mathrm{D}} \approx 1.8$–$2.4$) for filamentary structures at all epochs.

The observed continuous fractal spectrum, persistent low-dimensional channels, extreme thinness of high-redshift filaments (0.15–0.75 Mpc), north–south anisotropy, and galaxy-spin direction asymmetry are naturally explained if spacetime curvature emerges solely from the distribution and self-gravitating perturbations of baryonic matter. In this picture, dark matter and dark energy appear as unnecessary mathematical artifacts of enforcing global homogeneity on an intrinsically fractal spacetime.

We propose a testable, parameter-free framework in which dark components are replaced by an emergent, matter-dependent spacetime geometry. This approach simultaneously accounts for satellite planes, early massive galaxies, the Hubble tension, and the observed cosmic-web topology, while remaining fully consistent with the successes of ΛCDM on the largest scales ($D_{3\mathrm{D}} \to 3$ above ~ 300 Mpc). Numerical solutions within the framework (Appendix B) demonstrate that a fractal dimension $D_{3\mathrm{D}} \approx 2.25$ on ~ 10–30 Mpc scales can naturally produce a local Hubble constant ~ 5–$6\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ higher than the Planck-inferred value, offering a potential resolution to the Hubble tension without additional free parameters.

Upcoming surveys with Euclid DR1, the Nancy Grace Roman Space Telescope, and JWST Cycles 4–6 will provide decisive tests of the predicted scaling behavior, filamentary

dynamics, and redshift evolution of the fractal dimension. Confirmation of this trend would motivate a re-examination of structure formation mechanisms and open new avenues beyond the standard cold-dark-matter paradigm. If validated, this baryon-driven framework establishes a new cosmological paradigm in which spacetime itself is a direct, emergent extension of visible matter.

Recent empirical support for baryon-driven spacetime curvature, such as the quasi-periodic disk-jet co-precession observed in the tidal disruption event AT2020afhd (Li, 2025), further strengthens the viability of this approach. The model makes several quantitative, testable predictions, including scale-dependent variations in the Hubble parameter and persistently low fractal dimensions in future Euclid filament surveys.

A Preliminary High-Redshift Photometric Samples

Due to small sample sizes ($N = 12$ at $z > 15$ and $N = 7$ at $z > 20$), limited spectroscopic confirmation, and significant potential interloper contamination from dust-obscured or low-redshift sources, the photometric candidates at $z > 15$ and $z > 20$ are presented here strictly as exploratory results. The measured 3D fractal dimensions remain $D_{3\mathrm{D}} = 1.38 \pm 0.09$ ($z > 15$) and $D_{3\mathrm{D}} = 1.32 \pm 0.14$ ($z > 20$), following the same box-counting methodology as the main samples but with wider 95% confidence intervals driven by photometric redshift uncertainties ($\sigma_z/(1+z) \sim 0.05$–0.1) and deprojection assumptions.

Recent JWST analyses of high-redshift photometric samples (e.g., from JADES DR5, CEERS, PRIMER, NGDEEP, UNCOVER) confirm that robust $z \gtrsim 15$ candidates are scarce, with strong evidence for contamination by dusty star-forming or quiescent galaxies at $z \sim 3$–8 (e.g., Castellano, 2025; Witstok, 2026). Conservative Monte Carlo resampling incorporating contamination rates of 40–80% (spanning and extending literature estimates; Bisigello 2025; Naidu 2025; Zavala 2025) yields surviving $D_{3\mathrm{D}} \approx 1.35$ (95% CI: 1.19–1.52) even in the most pessimistic scenarios. These low values suggest extreme linearity and filamentary structure in the earliest biased environments, though full spectroscopic validation (via JWST Cycle 4–6 NIRSpec or Roman) remains essential.

The "Little Red Dots" (LRDs) population, prominently identified at $z \gtrsim 4$–10 and interpreted as accreting direct-collapse black holes (DCBHs) or dense-gas-enshrouded young SMBHs in overdense cocoons (Pacucci, 2026; Rusakov, 2026), provides a compelling analog at slightly lower redshifts. Their compact morphology (~ 100 pc scales), overmassive BH signatures, and association with pristine, low-metallicity gas environments align naturally with the enhanced gravitational efficiency in our low-dimensional ($D_{3\mathrm{D}} \lesssim 1.5$) baryonic channels at high z. While direct LRD analogs at $z > 15$–20 are not yet confirmed (due to rarity and contamination risks), the observed decreasing $D_{3\mathrm{D}}$ trend toward cosmic dawn supports the emergent model's prediction that baryon-driven perturbations foster efficient, filamentary collapse and early heavy seeding without dark halos. Future deeper surveys (Euclid DR1, Roman, JWST Cycles 4–6) will provide decisive tests.

B Exploratory Fractal-Gravity Framework

To illustrate that the observed fractal spectrum can arise within existing theoretical frameworks, we outline a fractional extension of general relativity in which the effective spacetime dimension $D(x)$ varies locally with the baryonic energy density, following the multi-fractional spacetime approach of Calcagni (2017, 2023).

B.1 Generalized Einstein–Hilbert Action

The standard Einstein–Hilbert action is

$$S = \frac{c^4}{16\pi G} \int d^4x \sqrt{-g}\, R + S_{\text{matter}}[g_{\mu\nu}, \psi],$$

where d^4x is the 4D Lebesgue measure and R is the Ricci scalar. In the fractional extension, the measure is replaced by a Hausdorff (fractional) measure $D^D x$ with dimension $D \in [3, 4]$, varying locally as $D(x)$. The action becomes

$$S = \frac{c^4}{16\pi G} \int D^D x \sqrt{-g}\, R + S_{\text{matter}}[g_{\mu\nu}, \psi].$$

For practical calculations, the measure can be expressed as $D^D x = v(x)\, d^4x$, where $v(x)$ encodes the fractional weight. Varying the action with respect to the metric yields modified Einstein equations of the form

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^4} T_{\mu\nu} + \Delta_{\mu\nu},$$

where $\Delta_{\mu\nu}$ encodes corrections due to the varying measure, which behave as an effective cosmological term proportional to deviations from $D = 4$.

B.2 Modified Friedmann Equations

Assuming a locally fractional FLRW metric with scale factor $a(t)$, the modified Friedmann equation reads

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda_{\text{eff}}}{3},$$

where

$$\Lambda_{\text{eff}}(x) = \alpha_\lambda \big(D(x) - 3\big) \rho_{\text{baryon}}(x).$$

The effective cosmological term Λ_{eff} is motivated by the positive pressure-like contribution arising when $D > 3$ in multi-fractional cosmologies. The constants α_λ, α, and β (see below) are positive and set by the characteristic curvature and density scales of baryonic structures. In high-density regions ($D \to 3$), perturbations grow faster; in low-density voids ($D \to 4$), an effective repulsive term emerges.

B.3 Dimension Evolution Equation

The dimension evolves according to the relaxation equation

$$\frac{\partial D}{\partial t} = \kappa \big(D_{\text{target}}(R) - D\big),$$

where $D_{\text{target}}(R) = 3 + \alpha \exp(-\beta R)$ is a monotonic function of the curvature scalar R, and $\kappa = \gamma H(t)$ with $\gamma \approx 1$ ties the relaxation rate to the Hubble timescale ($\kappa^{-1} \sim H^{-1} \approx$ 14 Gyr today).

B.4 Numerical Solutions and Stability

Simple numerical solutions (Runge–Kutta integration on baryon perturbation fields) yield fractal dimensions in the range $1.25 \leq D_{3\mathrm{D}} \leq 2.25$ in biased channels, consistent with the observational results presented in Section 3. Linear stability analysis around the equilibrium $D = D_{\mathrm{target}}$ gives $\delta D(t) = \delta D_0 e^{-\kappa t}$, which is stable for $\kappa > 0$. No ghosts or tachyonic instabilities arise in this classical extension.

Full cosmological simulations with CMB constraints are deferred to future work. Preliminary fits suggest $\gamma \sim 0.5$–2, compatible with BAO and supernova data. This framework offers a promising way to describe baryon-driven cosmic acceleration and large-scale structure formation. It aligns with the Unified Fractal Quantum Field Theory (UFQFT; Sogukpinar 2025) and is compatible with Verlinde's emergent gravity (Verlinde, 2017, 2023), where entropic forces from baryonic entanglement create effective repulsion in low-density regions. It can be further strengthened with multi-fractional quantum gravity models (Calcagni, 2025).

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Data and Code Availability

All analysis scripts, processed data, and numerical notebooks are publicly available at `https://github.com/janoscsabakeves-afk/emergent-fractal-spacetime.git` A permanent Zenodo archive with DOI will be created upon publication.

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